Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Optical clocks have reached fractional systematic uncertainties near 10^{-19} and, in separate lattice-clock experiments, instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is averaging time.[1,2] Connecting these instruments has extended precision frequency measurements from individual laboratories to optical-clock networks.[3,4,5] Optical fibres provide terrestrial connections, while long-distance free-space experiments are developing the transfer capability needed for future ground-to-space links.[6,7] These networks support the development of optical timekeeping and a future redefinition of the second, while providing a way to compare the gravitational environments of distant sites.[8,9]

Clocks sense gravitational potential through their relative ticking rates: a fractional frequency change of 10^{-18} corresponds to about 1 cm of equivalent height near Earth's surface.[9] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[10,11,12,13] A more demanding goal is to recover a changing potential and distinguish its physical contributions under realistic network operation. Closely spaced periodic signals require sufficient phase coverage, whereas irregular changes require precision within the duration of the event. Interruptions can also leave the frequency reference level uncertain between observing runs. Establishing how sensitivity, observation duration and continuity determine the accessible signals is therefore central to using clock networks as instruments for the dynamic Earth.

Tides provide a predictable signal against which to test this capability. The Moon's and Sun's tidal fields combine with deformation of the solid Earth and ocean mass redistribution. Solid-Earth tidal potentials can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$ at individual sites; smaller effects from pole tides and non-tidal mass changes in the atmosphere, oceans and continental water can matter at the 10^{-18} fractional-frequency level.[14] Solar heating also drives atmospheric pressure tides that force the ocean, while polar motion generates a slower pole tide through changes in centrifugal forcing.[15,16] Beyond these periodic signals, satellite gravity has revealed mass redistribution associated with the changing water cycle and climate.[17,18] Clock networks could provide repeated measurements between fixed sites to investigate such changes.[19] Their observable signal is the spatial difference, whose strength and time dependence depend on the baseline and the process being studied.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan, separated by 692 km and connected by a 1,350-km fibre link.[20] The combined solid-Earth and ocean-loading prediction gives an M_2 fractional-frequency amplitude of 4.78×10^{-18}, equivalent to about 4.4 cm of height.[16,21] We use 545.9 hours of observations spread over 353 days. Astronomical phase remains known through interruptions, allowing each observing segment its own frequency offset. The changing relative phase of the lunar and solar tides also permits joint estimation of the M_2+N_2 and neighbouring S_2+K_2 responses, with phases and relative amplitudes fixed within each group. The experiment measures a time-varying potential difference and tests the separation of neighbouring semidiurnal responses within one interrupted clock record.

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